Aarav Verma

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EDUCATION

University of Illinois at Urbana-Champaign May 2029
Grainger College of Engineering - Bachelor of Science in Aerospace Engineering with a Minor in Computer Science

SKILLS

CAD & Design: SolidWorks, Fusion, Siemens NX, Onshape, GD&T (ASME Y14.5), 3D Printing (PLA, ABS)
Simulation & Analysis: ANSYS FEA, ANSYS Fluent, MATLAB (PDE Toolbox), Simulink
Programming & Software: MATLAB, Python (NumPy, Matplotlib, PyQt5)

WORK EXPERIENCE

Tarform - Champaign, IL Aug 2026 - Current
Mechanical Design Engineer Intern (Part-Time)

- Create and maintain production documentation, including manufacturing drawings, for motorcycle components in SolidWorks, applying GD&T per ASME Y14.5 for machining and fabrication
- Prepare manufacturing drawings and GD&T tolerance views for a back rack assembly (LH/RH side structures, rear structure, aluminum plate) in SolidWorks, ensuring proper fit and assembly across all components

LEADERSHIP & INVOLVEMENT

National Organization for Business and Engineering - Champaign, IL Feb 2026 - Current
Project Manager

- Lead a cross-functional engineering team in developing a palm-sized autonomous interceptor drone for Haylon Technologies, integrating a proprietary battery management system into a compact quad-rotor airframe
- Review CAD models, FEA simulations, and MATLAB validation calculations from developers, and coordinate PID controller tuning to achieve reliable in-flight tracking ahead of prototyping and assembly
- Design the architecture for the drone's vision-based tracking system, using camera input to detect and intercept airborne objects, and review developer code to keep implementation aligned with that design

Engineering Lead, Canary Drone Project Feb 2026 - May 2026

- Redesigned propeller to toroidal configuration in Fusion, simulated (ANSYS Fluent) improved thrust efficiency ~11% at 4-6k RPM and validating structural integrity via FEA (Factor of Safety > 2.0) under 12N thrust loads
- Designed a modular drone airframe in Fusion with a Jetson housing frame and tool-free foldable leg assembly, presented at Cozad New Venture Challenge against 300+ teams for a $500,000 prize pool

Ghost Electric Motorcycles - Champaign, IL Aug 2025 - Jan 2026
Drivetrain Engineer

- Engineered a load-bearing drivetrain frame and rear suspension swingarm in SolidWorks, reverse-engineering the 2010 Kawasaki ZX-6R to replicate OEM anti-squat ratio and pivot geometry within ±0.05 mm
- Validated drivetrain stiffness and load capacity in ANSYS FEA and MATLAB under peak torque delivery and chain tension across braking, cornering, and acceleration loading conditions

PROJECTS

Actuator-Limited Flight Controller - Champaign, IL Jul 2026 - Sep 2026
Simulink

- Designed a cascaded rate/angle PID controller (Betaflight/PX4-style) for a multi-axis drone model, cutting peak overshoot from 40% to 10% by diagnosing and correcting a derivative-kick and noise-amplification problem
- Characterized actuator saturation and integral anti-windup behavior by sweeping the motor torque limit to locate the boundary between recoverable windup and an unrecoverable, underpowered-actuator failure mode
- Built a MATLAB 3D visualization pipeline that animates drone attitude from logged Simulink flight data using composed rotation matrices, verifying transformation correctness through direct numerical checks

Parametric Airfoil Structural Analysis - Olympia, WA Aug 2026

- Generated NACA 0012 airfoil geometry in MATLAB, modeled the solid in Fusion, and imported STEP geometry into MATLAB for structural FEA under directional surface loads
- Designed and ran a mesh convergence experiment to test sensitivity of stress results to mesh refinement finding a stress singularity, added a CAD fillet that brought successive mesh results within 2% of each other

Cantilever Bracket FEA - Olympia, WA Aug 2026

- Designed a hand-calc vs. FEA comparison experiment to validate tip deflection and bending stress predictions, catching a 10x error in the FEA load application by querying the true load-face area from the solved mesh
- Reconciled corrected FEA output to within 10% of hand calculations, attributing the remaining gap to bend-fillet compliance and mounting-hole stress concentrations the beam model omits